

UNIT-2 AGILE MINI-PROJECT

Enterprise Student Record and Academic Management Platform

Student Name	Krushna Dattatray Dhokane
PRN	2125UDSM1116
Roll No.	116
Project	Student Record and Academic Management Platform
Methodology	Agile / Scrum

Academic Submission – Unit 2

Contents

1. Introduction and Agile Project Planning
2. Agile Product Vision
3. Scrum Team Structure
4. Scrum Roles and Responsibilities
5. Epic Documentation
6. User Story Repository
7. Acceptance Criteria
8. Prioritized Product Backlog
9. Story Point Estimation
10. Sprint Planning Document
11. Sprint Backlog
12. Scrum Board
13. Scrum Ceremonies
14. Burndown Chart
15. Sprint Review Report
16. Sprint Retrospective Report
17. Agile Framework Comparison
18. Project Timeline and Release Roadmap
19. Definition of Done
20. Unit-2 Deliverables Checklist

Note: The structure and terminology follow the Unit-2 assignment brief. Project-specific stories, priorities, estimates and reports are prepared as an academic Agile planning draft based on that brief.

1. Introduction and Agile Project Planning

Unit 2 converts the Student Record and Academic Management Platform identified during product discovery into an Agile development plan. The platform is intended to manage student records and academic information in a higher-education environment.

Project objective

- Translate the product vision into Agile deliverables.
- Organize functionality into Epics and User Stories.
- Prioritize the Product Backlog using business value, user importance, dependencies and complexity.
- Plan incremental delivery through Sprints.
- Use Scrum roles, ceremonies and a Scrum Board to monitor progress.

Project scope

- Student Management
- Course Management
- Attendance Management
- Marks and Academic Performance
- Authentication and Security
- Notifications
- Reporting

2. Agile Product Vision

Product Vision

Develop a secure and user-friendly Student Record and Academic Management Platform that enables educational institutions to efficiently manage student records, attendance, academic performance, course registration, and reporting through a scalable web-based application.

Primary users

- Students
- Faculty Members
- Academic Administrators
- Examination Cell Staff
- IT Support Personnel

Expected outcomes and success criteria

- Centralized academic information.
- Reduced dependence on manual record handling.
- Accurate maintenance of student and academic records.
- Easier generation and review of academic reports.
- Secure access for authorized users.
- Incremental delivery of core modules through Sprints.

Project constraints

The Unit-2 brief asks students to identify project constraints. For this academic draft, constraints should be reviewed by the team during Sprint Planning rather than assumed as fixed technical constraints.

3. Scrum Team Structure

Role	Main Responsibility
Product Owner	Defines product requirements, prioritizes the Product Backlog, represents stakeholder interests and accepts completed work.
Scrum Master	Facilitates Scrum ceremonies, removes project impediments, supports collaboration and ensures Agile practices are followed.
Development Team	Designs the application, writes code, tests software, prepares documentation and demonstrates completed features.

4. Scrum Roles and Responsibilities

Product Owner

Owns product value, maintains and prioritizes the Product Backlog, and accepts completed work.

Scrum Master

Facilitates Scrum events, helps remove impediments and supports effective Agile practices.

Development Team

Builds and tests increments and prepares documentation and demonstrations.

5. Epic Documentation

Epic 1 – Student Management

Purpose and scope: Organize related functionality into a high-level functional module.

- Student Registration
- Student Profile Management
- Student Search
- Student Update
- Student Deletion

Epic 2 – Course Management

Purpose and scope: Organize related functionality into a high-level functional module.

- Add Course
- Update Course
- Assign Students
- View Course Details

Epic 3 – Attendance Management

Purpose and scope: Organize related functionality into a high-level functional module.

- Record Attendance
- View Attendance
- Attendance Reports

Epic 4 – Marks and Academic Performance

Purpose and scope: Organize related functionality into a high-level functional module.

- Enter Marks
- Update Marks
- Calculate Grades
- Generate Result Sheets

Epic 5 – Authentication and Security

Purpose and scope: Organize related functionality into a high-level functional module.

- User Login
- Logout
- Password Reset
- Role-Based Access Control

Epic 6 – Notifications

Purpose and scope: Organize related functionality into a high-level functional module.

- Attendance Alerts
- Examination Notifications
- Result Notifications

Epic 7 – Reporting

Purpose and scope: Organize related functionality into a high-level functional module.

- Student Reports
- Attendance Reports
- Performance Reports
- Course Reports

6. User Story Repository

User stories use the format: As a [user], I want to [action], so that [benefit]. The Unit-2 brief requires stories for the identified features.

ID	Epic	User Story	Priority	SP
US-01	Authentication	As a registered user, I want to log in so that I can securely access the platform.	High	3
US-02	Student Management	As an administrator, I want to add a student so that the student's academic record can be maintained.	High	5
US-03	Student Management	As an administrator, I want to search for a student so that I can quickly find a record.	High	3
US-04	Student Management	As an administrator, I want to update a student record so that information remains current.	Medium	3
US-05	Course Management	As an administrator, I want to add a course so that course information is available.	Medium	3
US-06	Attendance Management	As a faculty member, I want to record student attendance so that attendance records remain accurate and updated.	High	5

ID	Epic	User Story	Priority	SP
US-07	Attendance Management	As a faculty member, I want to view attendance so that I can monitor student participation.	Medium	3
US-08	Marks & Academic Performance	As a faculty member, I want to enter marks so that academic performance can be recorded.	High	5
US-09	Marks & Academic Performance	As a system user, I want grades to be calculated so that results can be interpreted consistently.	Medium	5
US-10	Notifications	As a student, I want attendance and examination notifications so that I do not miss important academic information.	Medium	5
US-11	Reporting	As an administrator, I want attendance reports so that academic records can be reviewed.	Medium	5
US-12	Reporting	As an administrator, I want performance reports so that student academic performance can be reviewed.	Medium	5

7. Acceptance Criteria

Acceptance criteria are measurable conditions that determine whether a User Story is complete. The Unit-2 brief specifically illustrates criteria for Student Login and Faculty Attendance.

US-01 – Authentication

User Story: As a registered user, I want to log in so that I can securely access the platform.

- Login page loads successfully.
- Registered users can log in.
- Invalid credentials display an appropriate error.
- Successful login redirects to the dashboard.
- Session management functions correctly.

US-02 – Student Management

User Story: As an administrator, I want to add a student so that the student's academic record can be maintained.

- Required student details can be entered.
- Input is validated.
- The student record is saved successfully.

US-03 – Student Management

User Story: As an administrator, I want to search for a student so that I can quickly find a record.

- Search accepts relevant student information.
- Matching student records are displayed accurately.

US-04 – Student Management

User Story: As an administrator, I want to update a student record so that information remains current.

- An existing record can be selected.
- Details can be edited and validated.
- Updated information is saved successfully.

US-05 – Course Management

User Story: As an administrator, I want to add a course so that course information is available.

- Course details can be entered.
- Course information is validated.
- The course is saved successfully.

US-06 – Attendance Management

User Story: As a faculty member, I want to record student attendance so that attendance records remain accurate and updated.

- Faculty can select a course.
- Faculty can select a lecture date.
- Faculty can mark attendance.

- Attendance records are saved successfully.
- Attendance reports can be generated.

US-07 – Attendance Management

User Story: As a faculty member, I want to view attendance so that I can monitor student participation.

- Attendance for the selected course/date is displayed correctly.
- The displayed records match saved attendance data.

US-08 – Marks & Academic Performance

User Story: As a faculty member, I want to enter marks so that academic performance can be recorded.

- Marks can be entered for the correct student/course.
- Marks are validated.
- The record is saved successfully.

US-09 – Marks & Academic Performance

User Story: As a system user, I want grades to be calculated so that results can be interpreted consistently.

- Grade calculation uses defined result rules.
- The calculated grade is displayed correctly.

US-10 – Notifications

User Story: As a student, I want attendance and examination notifications so that I do not miss important academic information.

- Relevant notifications are generated.
- Notifications are delivered to intended users.

US-11 – Reporting

User Story: As an administrator, I want attendance reports so that academic records can be reviewed.

- Attendance report can be generated.
- Report contains accurate attendance records.

US-12 – Reporting

User Story: As an administrator, I want performance reports so that student academic performance can be reviewed.

- Performance report can be generated.
- Report summarizes the correct academic data.

8. Prioritized Product Backlog

Prioritization follows the factors stated in the brief: business value, user importance, technical dependencies and implementation complexity.

Priority	Epic	User Story	SP
High	Authentication	US-01 – As a registered user, I want to log in so that I can securely access the platform.	3
High	Student Management	US-02 – As an administrator, I want to add a student so that the student's academic record can be maintained.	5
High	Student Management	US-03 – As an administrator, I want to search for a student so that I can quickly find a record.	3
High	Attendance Management	US-06 – As a faculty member, I want to record student attendance so that attendance records remain accurate and updated.	5
High	Marks & Academic Performance	US-08 – As a faculty member, I want to enter marks so that academic performance can be recorded.	5
Medium	Student Management	US-04 – As an administrator, I want to update a student record so that information remains current.	3
Medium	Course Management	US-05 – As an administrator, I want to add a course so that course information is available.	3
Medium	Attendance Management	US-07 – As a faculty member, I want to view attendance so that I can monitor student participation.	3
Medium	Marks & Academic Performance	US-09 – As a system user, I want grades to be calculated so that results can be interpreted consistently.	5
Medium	Notifications	US-10 – As a student, I want attendance and examination notifications so that I do not miss important academic information.	5
Medium	Reporting	US-11 – As an administrator, I want attendance reports so that academic records can be reviewed.	5
Medium	Reporting	US-12 – As an administrator, I want performance reports so that student academic performance can be reviewed.	5

9. Story Point Estimation

Points	Complexity
1	Very Easy
2	Easy
3	Moderate
5	Difficult
8	Very Difficult
13	Highly Complex

The estimates in this academic draft are relative planning estimates and can be refined collaboratively during Planning Poker and Sprint Planning.

10. Sprint Planning Document

Sprint 1

Goal: Develop user authentication and project setup.

Activities:

- Repository initialization
- User registration
- Login functionality
- Basic user interface
- Testing

Sprint 2

Goal: Develop Student Management Module.

Activities:

- Add Student
- Search Student
- Update Student
- Delete Student

Sprint 3

Goal: Develop Course and Attendance Management.

Activities:

- Add Course
- Assign Courses
- Record Attendance
- Attendance Reports

11. Sprint Backlog

Sprint	Backlog Items	Expected Increment
Sprint 1	Authentication, repository setup, registration, login, basic UI, testing	Working authentication and initial project setup
Sprint 2	Add, search, update and delete student records	Student Management module increment
Sprint 3	Add course, assign courses, record attendance, attendance reports	Course and Attendance Management increment

12. Scrum Board

The assignment brief specifies the workflow: Backlog → To Do → In Progress → Testing → Completed.

Backlog	To Do	In Progress	Testing	Completed
Login	Add Student	Search Student	Reports	Setup
Course	Attendance	Notifications		

13. Scrum Ceremonies

Sprint Planning

Select User Stories, estimate effort and define the Sprint Goal.

Daily Scrum

Each member answers: What did I complete yesterday? What will I do today? What challenges am I facing?

Sprint Review

Demonstrate completed software features and collect stakeholder feedback.

Sprint Retrospective

Discuss what went well, what challenges occurred and what improvements should be implemented.

14. Burndown Chart

A burndown chart tracks remaining work across a Sprint. The following is a planning template using the estimated Sprint workload; actual values should be updated during execution.

Sprint Day	Planned Remaining Story Points	Actual Remaining Story Points
Day 0	13	Update during Sprint
Day 1	10	Update during Sprint
Day 2	7	Update during Sprint
Day 3	5	Update during Sprint
Day 4	3	Update during Sprint
Day 5	1	Update during Sprint
Day 6	0	Update during Sprint

Note: The Unit-2 brief requires a Burndown Chart but does not prescribe specific day-by-day values. Therefore, the values above are a clearly labelled planning template, not reported project results.

15. Sprint Review Report

Sprint 1

Review focus: Develop user authentication and project setup.

Demonstration items: Repository initialization, User registration, Login functionality, Basic user interface, Testing.

Stakeholder feedback: Record feedback received during the review and convert accepted changes into Product Backlog items.

Sprint 2

Review focus: Develop Student Management Module.

Demonstration items: Add Student, Search Student, Update Student, Delete Student.

Stakeholder feedback: Record feedback received during the review and convert accepted changes into Product Backlog items.

Sprint 3

Review focus: Develop Course and Attendance Management.

Demonstration items: Add Course, Assign Courses, Record Attendance, Attendance Reports.

Stakeholder feedback: Record feedback received during the review and convert accepted changes into Product Backlog items.

16. Sprint Retrospective Report

What went well

Completed planned stories, maintained a visible backlog and used regular Scrum communication.

What challenges occurred

Potential issues may include unclear requirements, dependencies, testing delays or changing priorities.

What should improve

Clarify acceptance criteria early, estimate collaboratively, surface blockers in Daily Scrum and test continuously.

17. Agile Framework Comparison

Framework	Focus	Suitability for this project
Scrum	Defined roles, backlog, sprints and ceremonies	Recommended for this project because the Unit-2 brief is organized around Scrum roles, backlog and Sprints.
Kanban	Continuous flow and visual work management	Useful where priorities change frequently.
Extreme Programming (XP)	Engineering practices and frequent feedback	Useful when technical quality and rapid feedback are dominant needs.
Lean Development	Eliminating waste and maximizing value	Useful for efficiency-focused delivery.
Large Scale Scrum (LeSS)	Scaling Scrum across multiple teams	More suitable for larger multi-team products.
SAFe	Structured scaling for large organizations	More suitable for large, complex enterprise portfolios.
Scrum@Scale	Coordination of multiple Scrum teams	Useful when several Scrum teams need coordinated delivery.

Recommended framework

Scrum is recommended for this academic project because the Unit-2 brief explicitly organizes the work around Scrum roles, a Product Backlog, Sprints, a Scrum Board and Scrum ceremonies.

18. Project Timeline and Release Roadmap

Phase	Planned Work	Release / Outcome
Phase 1	Review Unit 1 product discovery, refine Agile Product Vision, organize Scrum Team and repository	Agile planning baseline
Sprint 1	Authentication and project setup	Initial working increment
Sprint 2	Student Management	Student Management increment
Sprint 3	Course and Attendance Management	Course + Attendance increment
Future Increment	Marks, Notifications, Reporting and refinement from review feedback	Expanded academic platform

19. Definition of Done

- Acceptance criteria are satisfied.
- Implementation is tested.
- Required documentation is prepared.
- The feature can be demonstrated during the Sprint Review.
- For Student Login specifically: the page loads, registered users can log in, invalid credentials show appropriate errors, successful login redirects to the dashboard and session management functions correctly.

20. Unit-2 Deliverables Checklist

No.	Deliverable	Included in this PDF
1	Agile Project Planning Report	Yes – see relevant section
2	Product Vision Document	Yes – see relevant section
3	Scrum Team Structure	Yes – see relevant section
4	Scrum Roles and Responsibilities Document	Yes – see relevant section
5	Epic Documentation	Yes – see relevant section
6	User Story Repository	Yes – see relevant section
7	Acceptance Criteria Document	Yes – see relevant section
8	Prioritized Product Backlog	Yes – see relevant section
9	Story Point Estimation Sheet	Yes – see relevant section
10	Sprint Planning Document	Yes – see relevant section
11	Sprint Backlog	Yes – see relevant section
12	Scrum Board	Yes – see relevant section
13	Burndown Chart	Yes – see relevant section
14	Sprint Review Report	Yes – see relevant section
15	Sprint Retrospective Report	Yes – see relevant section
16	Agile Framework Comparison Report	Yes – see relevant section
17	Project Timeline and Release Roadmap	Yes – see relevant section
18	Complete Agile Documentation	Yes – see relevant section

End of Unit-2 Agile Mini-Project